



Seminar ON Management Of Shock In Children

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Outline

- Objectives
- Introduction & Definition
- Pathophysiology
- Stages of Shock
- Classification of Shock
- Clinical Presentations
- Principles of Management
- Specific Management by Type
- Prognosis & Summary

Objective

At the end of this seminar the participants will be able to;

- Define shock and describe its pathophysiology
- Identify the stages and classifications of pediatric shock
- Recognize clinical presentations and diagnostic indicators
- Outline evidence-based management strategies
- Describe complications and prognosis of shock

Definition & Significance

- **Definition:** An acute, life-threatening state of cardiovascular insufficiency characterized by inadequate delivery of oxygen and nutrients to meet the metabolic demands of vital organs, resulting in cellular dysfunction and injury.
- Shock results from energy failure due to inadequate oxygen and glucose delivery to meet metabolic demands. It is a leading cause of death and disability in children worldwide.

Epidemiology

- Shock occurs in approximately 2% of all hospitalized infants, children, and adults in the USA, and the mortality rate varies according to the clinical circumstances.
- Many die because of complications and Multiple Organ Dysfunction Syndrome (MODS) .
- Early recognition and treatment significantly improve outcomes.

PATHOGENESIS OF SHOCK

Inadequate tissue
perfusion

Tissue hypoxia



Anaerobic metabolism



Accumulation of lactic
acid & CO₂



Irreversible cell and
organ damage

Stages of Shock

- Shock progresses along a continuum of severity :

Compensated Shock:

- Compensatory mechanisms maintain adequate blood pressure
- Tachycardia and increased systemic vascular resistance occur
- Blood pressure remains normal despite inadequate tissue perfusion
- A child can lose 30-35% of circulating blood volume and still maintain normal blood pressure

Stages of Shock

Decompensated (Hypotensive) Shock:

- Compensatory mechanisms fail
- Hypotension develops (blood pressure drops below age-specific thresholds)
- End-organ perfusion is severely compromised

Irreversible Shock:

- Progressive end-organ dysfunction and failure occur
- Cellular death follows
- The patient is unresponsive to resuscitative efforts

Critical Takeaway: Hypotension is a late finding in pediatric

Classification of Shock

1. Hypovolemic shock
2. Cardiogenic shock
3. Distributive shock: septic shock, Anaphylactic shock, Neurogenic shock
4. Obstructive shock: Cardiac tamponade, Tension pneumothorax, Massive pulmonary embolism.

Classification of Shock

- Shock is generally categorized into five major types :
- **Hypovolemic Shock:**

This is the most common cause of shock in children worldwide. It is most frequently caused by acute fluid losses resulting from diarrhoea, vomiting, or haemorrhage.

Classification of Shock

- **Distributive Shock:**

This type is characterized by inappropriate vasodilation, pooling of blood in the peripheral vasculature, and leakage of blood plasma through a compromised endothelial barrier.

Common causes include:

- sepsis
- anaphylaxis
- neurogenic shock and drug overdose.

Classification of Shock

Cardiogenic Shock:

This results from pump failure and decreased cardiac output. It is seen in patients with:

- congenital heart disease
- congenital or acquired cardiomyopathies
- acute myocarditis, and multisystem inflammatory syndrome in children (MIS-C).

Classification of Shock

Obstructive Shock:

This occurs when a lesion creates a mechanical impediment to ventricular blood outflow. Causes include:

- pericardial tamponade
- tension pneumothorax, pulmonary embolism and ductal-dependent left-sided heart lesions.

Classification of Shock

- **Dissociative Shock:**

This is a special category in which cellular oxygen delivery or utilization is severely impaired despite sustained or supernormal tissue perfusion. Causes include severe anaemia, methemoglobinemia, carbon monoxide poisoning, and thiamine deficiency.

- **Note:** Septic shock is often discussed in conjunction with distributive shock but involves a more complex interaction of distributive, hypovolemic, and cardiogenic components

Septic shock

- fever (or hypothermia) along with evidence for either a systemic or localized infection
- Infants and young children most commonly present with “cold shock
- Older children and adolescents more commonly present with “warm shock

Neurogenic shock

- is a type of distributive shock
- caused by a sudden disruption of sympathetic nerve stimulation to the vascular smooth muscle leading to a profound decrease in vasomotor tone,
- Can results in bradycardia, or at least absence of the usual tachycardic response to hypotension.

Table 85.1 Types of Shock

TYPES OF SHOCK	PATHOPHYSIOLOGY	APPEARANCE	EXAMPLES	TREATMENTS
Hypovolemic*	Decreased cardiac output due to volume depletion → decreased preload	Tachycardia and vasoconstriction maintain adequate circulation up to 30% of circulating volume Narrowed pulse pressure, delayed capillary refill, orthostatic changes, late hypotension, AMS, and decreased UOP	Nonhemorrhagic (vomiting and diarrhea) Hemorrhagic (trauma)	IV fluid bolus Vasopressors Blood replacement
Cardiogenic	Decreased cardiac output due to myocardial dysfunction, increased afterload, and/or lack of ventricular filling	Tachycardia, vasoconstriction, cool extremities, narrow pulse pressure, delayed capillary refill, respiratory distress, rales or gallop rhythm, enlarged liver, JVD, cardiomegaly on chest radiograph Significant gradient between upper and lower extremity blood pressures	Cardiomyopathies, infectious myocarditis, and systemic inflammatory process, autoimmune disease, impaired coronary perfusion, cardiopulmonary bypass, acidosis, HIE, and dysrhythmias Infants: ductal-dependent lesions, tachydysrhythmias	Judicious IV fluid resuscitation, inotropic agents to improve contractility, vasodilators to reduce afterload, and management of tachyarrhythmias PGE for infants <2 mo

Distributive	Decreased cardiac output and systemic vascular resistance due to peripheral vasodilation → decreased afterload and preload, redistribution of blood flow away from vital organs, and loss of sympathetic outflow	Tachycardic, vasodilated, flushed, warm extremities, wide pulse pressure, bounding pulses, flash capillary refill Anaphylaxis: rash; facial swelling; lip, tongue, or airway swelling; bronchospasm; hypotension Spinal shock: unable to raise HR, hypotension Neurogenic: Control ↑ ICP	Septic shock Toxic ingestions Anaphylaxis Spinal cord injuries	IV fluids, vasopressors, antibiotics Specific antidotes Remove triggers, IV fluids, IM epinephrine, antihistamines, vasopressors IV fluids, vasoconstrictors
Obstructive	Decreased cardiac output due to increased afterload of the right ventricle from an obstructive process	Tachycardia, delayed capillary refill, cool extremities, narrow pulse pressure, distended neck veins, distant heart tones, asymmetric breath sounds	Tension pneumothorax Pulmonary embolism Cardiac tamponade	Anticoagulants Drain pericardial effusion Evacuate pneumothorax
Dissociative	High cardiac output failure due to inadequate oxygen-releasing capacity	Tachycardia ± delayed capillary refill, cool extremities, weakened pulse, pallor or flushed cheeks	Anemia Carbon monoxide Methemoglobinemia Cyanide	Gradual fluid replacement and blood replacement 5 mL/dL Oxygen, hyperbaric chamber Methylene blue Hydroxocobalamin and cyanide antidote kit (sodium thiosulfate, sodium nitrate, amyl nitrite) plus oxygen

Shock					
Type	Hypovolemic	Cardiogenic	Obstructive	Distributive	Dissociative
Etiology	Major trauma Gastrointestinal bleed Dehydration	Myocardial infarction Myocarditis Arrhythmia	Pulmonary embolus Cardiac tamponade Tension pneumothorax Status asthmaticus	Sepsis Spinal cord injury Anaphylaxis Ingestion Neurogenic	Anemia, carbon monoxide or cyanide poisoning, methemoglobinemia*
Cardiovascular findings					
Preload (filling pressures)	↓	↑	↕	↓	↕
Cardiac contractility	↑	↓	↕	↑	↑
Cardiac output	↓	↓	↓	↕	↑
Afterload (peripheral tone)	↑	↑	↑	↓	↕
Hemodynamic support					
Relieve obstruction			+++		
Volume expansion	+++	+	+	+++	+
Inotropes		+++	+	+	
Vasopressors	+	+	+	++	

Fig. 85.1 The causes, cardiovascular findings, and hemodynamic support for different types of shock. Under cardiovascular findings, *bidirectional arrows* indicate variation in findings among patients with the particular type. Under hemodynamic support, the number of + signs indicates the importance of therapy. A combined + and – indicates that the intervention could help some patients but must be used with caution. *Treat specific etiology transfusion or specific antidotes. (Modified from Angus DC. Approach to the patient with shock. In: Goldman L, Schafer AI, eds. Goldman-Cecil Medicine, 26th ed. Philadelphia: Elsevier; 2020:642, Fig. 98-1.)

Classification Based on Treatment Response

Fluid-Responsive Shock:

- The patient shows hemodynamic improvement after initial fluid bolus (10-20 mL/kg)
- Heart rate decreases, perfusion improves
- No further escalation of therapy required

Fluid-Refractory Shock:

- Shock persists despite adequate fluid resuscitation
- Defined as requiring ≥ 40 -60 mL/kg of fluid without clinical improvement
- Requires initiation of vasoactive medications

Classification Based on Treatment Response

Catecholamine-Resistant Shock:

- Shock persists despite adequate doses of vasoactive agents
- Requires evaluation for atypical etiologies
- May need stress-dose corticosteroids or ECMO

Clinical Presentations

Signs and symptoms reflect impaired circulation and end-organ perfusion:

Vital Signs:

- Tachycardia is the most common early sign
- Tachypnea often occurs secondary to acidemia
- Bradycardia is a late and ominous sign

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Perfusion:

- Prolonged capillary refill time (>2 seconds)
- Diminished peripheral and/or central pulses
- Cool extremities and mottled skin
- Flash capillary refill and bounding pulses in distributive shock

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Mental Status:

- Irritability, confusion, lethargy, or restlessness
- Altered consciousness is an early sign in children

Other Signs:

- Oliguria (urine output <0.5 -1 mL/kg/hour)
- Dry mucous membranes and poor skin turgor
- Ileus and cyanosis
- Respiratory distress and diminished breath sounds
- Jugular vein distention and muffled heart sounds

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Diagnostic Indicators:

- Metabolic acidosis
- Hyperlactatemia (lactate $>2-4$ mmol/L is concerning)
- Low central venous oxygen saturation

General Principles of Management

- **Early Recognition:**

This is the key to survival. The initial signs of shock may be subtle in children as their compensatory mechanisms are very effective.

- **ABCDE Approach:**

- **A (Airway):** Ensure airway patency and adequacy of breathing. Consider intubation if the patient cannot protect their airway or if respiratory failure is present.
- **B (Breathing):** Provide supplemental oxygen if hypoxemia ($\text{SpO}_2 < 92-94\%$) is present. High-flow oxygen should be given to maintain SpO_2 above 95%.

General Principles of Management

- **C (Circulation):** Establish vascular access immediately (two large-bore IVs or intraosseous access if IV access is not achieved within 2 attempts).
- **D (Disability):** Assess neurological status and level of consciousness.
- **E (Exposure):** Check for other injuries, rashes, or sources of infection, and maintain normothermia.

General Principles of Management

Laboratory Evaluation:

- Blood culture (before antibiotics if possible)
- Blood gas analysis (including lactate)
- Blood glucose, electrolytes, calcium, magnesium
- Full blood count, coagulation studies

Antibiotics:

- In septic shock, broad-spectrum antibiotics should be given **within 1 hour** of recognition, ideally after blood cultures are obtained

Key Principle: Reassess the patient after every intervention to determine the response to therapy.

Fluid Resuscitation

Initial Bolus:

- 10-20 mL/kg isotonic crystalloid (Normal Saline or Ringer's Lactate) over 15-20 minutes
- Balanced salt solution is preferred over Normal Saline as bolus fluid therapy
- For cardiogenic shock: use 5-10 mL/kg boluses with careful monitoring

Subsequent Boluses:

- Repeat 10-20 mL/kg, reassessing after each
- Up to 40-60 mL/kg may be needed in the first hour
- May exceed 200 mL/kg over several hours in severe cases.

Fluid Resuscitation

Precautions:

- Assess the response of prior bolus therapy before repeating
- Ensure signs of fluid overload have not developed (hepatomegaly, crackles)
- Cumulative fluid overload $>10\%$ is associated with increased mortality

Escalation:

- For children with evidence of fluid overload, adequate intravascular volume, or lack of hemodynamic improvement with prior fluid boluses, resuscitation should shift from volume resuscitation to vasoactive medications.

Principles of Management - Vasoactive Medications

- **Indications:**

Vasoactive medications should be initiated for children with fluid-refractory shock.

Choice of Agent Based on Shock Type:

- **Epinephrine:**

This is the preferred first-line vasoactive therapy for fluid-refractory septic shock and cardiogenic shock. Epinephrine is preferred over dopamine as a first-line vasoactive in paediatric septic shock.

- **Norepinephrine:**

This is recommended as first-line vasoactive therapy for children with distributive shock or neurogenic shock (warm shock with low SVR).

Principles of Management - Airway and Breathing

Oxygen Therapy:

- Supplemental oxygen if SpO₂ is 92-94%
- Universal administration of high FiO₂ in severe anemia or dissociative shock
- All children with shock should receive high-flow oxygen to maintain SpO₂ above 95%

Indications for Intubation and Mechanical Ventilation:

- Type 1 respiratory failure (hypoxemic)
- Persistence of circulatory failure despite resuscitation
- Inability to protect the airway

Principles of Management - Airway and Breathing

- Depressed level of consciousness
- Impending respiratory failure

Important Consideration:

- Ensure fluid resuscitation is commenced prior to intubation, as anaesthetic agents and mechanical ventilation may cause hypotension in a shocked patient.

Sedation:

- Avoid etomidate for sedation during intubation in septic shock patients.

Principles of Management - Correcting Metabolic Abnormalities

Hypoglycemia:

- Always check blood glucose in any child with shock
- Hypoglycemia is common in shock patients
- If hypoglycemia is present (BGL <3.0 mmol/L or symptomatic), administer 2 mL/kg of 10% Glucose as a bolus
- Ensure maintenance fluids contain 5-10% Glucose

Principles of Management - Correcting Metabolic Abnormalities

Hypocalcemia:

- Calcium should be corrected during initial resuscitation
- Optimize serum calcium levels to support cardiac function

Electrolyte Abnormalities:

- Correct potassium and magnesium deficiencies
- These may optimize cardiac function

Acid-Base Disturbances:

- Metabolic acidosis is common and should be monitored
- Improving perfusion is the primary treatment for acidosis

Hemodynamic targets

Includes;-

1. Decrease in HR;
2. Increase in BP;
3. Normalization of capillary refill
4. Peripheral pulse strength, and

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5. Extremity temperature;

6. Improved urine output and

7. Fall in blood lactate by $\geq 10\text{--}20\%$.

- Before repeating a fluid bolus, assess the response of prior bolus therapy
- And ensure that signs of fluid overload have not developed.

Con'd

- For children with
 - 1) evidence of fluid overload,
 - 2) adequate intravascular volume,
 - 3) lack of hemodynamic improvement with prior fluid boluses,resuscitation should shift from volume resuscitation to vasoactive medications

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- The optimal fluid type for initial shock resuscitation could be

1. Crystalloids

0.9% normal saline

RL

Hartmann solution, and Plasma-Lyte

2. Colloids

Other treatments

- Resuscitation with blood products should be prioritized for children with hemorrhagic shock.
- Vasoactive medications should be initiated for children with fluid refractory shock,
- epinephrine is the preferred first-line vasoactive therapy for fluid-refractory septic shock

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- Epineprine is also the recommended initial vasoactive agent in children with cardiogenic shock.
- However, in children with a low-SVR state, distributive, septic, or neurogenic shock, norepinephrine as the first-line vasoactive therapy.
- hypoglycemia and hypocalcemia should be corrected during the initial resuscitation

Specific Management

Cardiogenic shock

- Early treatment with inotropes such as epinephrine, dopamine, dobutamine, or milrinone
- Cautious fluid administration (5-10 mL/kg over 20-30 minutes if needed)
- Avoid crystalloid infusion if pulmonary congestion is present
- Optimize serum calcium and potassium levels
- Correct hypoxia, rhythm disturbances, electrolyte abnormalities, and acid-base disturbances
- Consider cardiology consultation and echocardiography

Fluid administration

- Cautious fluid administration
- If Hypotension without pulmonary congestion ... a small fluid challenge
- If there is pulmonary congestion, crystalloid infusion should not be given

Vasodilators

- Vasodilators have been used to elevate cardiac output and to reduce left ventricular filling pressure.
- Risk: by lowering the already markedly reduced coronary perfusion pressure, myocardial perfusion can be compromised further.
- Milrinone is an inotrope/vasodilator which can augment contractility but can also decrease BP.

Other management

- Correct hypoxia, hypovolemia, rhythm disturbances, electrolyte abnormalities and acid-base disturbances
- Supplementary oxygen
- Antipyretics, analgesics, sedatives

Obstructive shock

- Action to relieve mechanical obstruction:
 - Thoracentesis for tension pneumothorax
 - Pericardiocentesis for cardiac tamponade
 - Systemic fibrinolytic medications for pulmonary embolism
 - Prostaglandin E1 infusion for neonates with ductal-dependent lesions

Anaphylactic Shock (Distributive):

- Airway and breathing management
- Fluid resuscitation
- Immediate administration of IM or IV epinephrine
- Epinephrine dose: 0.1 mL/kg of 1:10,000 solution every 3-5 minutes (maximum dose: 1 mg)
- If hypotension is refractory, start epinephrine infusion

Hypovolemic Shock:

- Aggressive fluid resuscitation
- Blood products for hemorrhagic shock

Septic shock

Septic shock

According to the International Pediatric Sepsis Consensus Conference and Nelson Textbook of Pediatrics, septic shock is defined as:

- Septic shock = Severe Sepsis + Cardiovascular Dysfunction
- Specifically, septic shock is:
- Sepsis (SIRS in the presence of suspected or proven infection)
PLUS
- Cardiovascular organ dysfunction characterized by:
- Hypotension (blood pressure <5th percentile for age) despite administration of isotonic intravenous fluid bolus ≥ 40 mL/kg in 1 hour, OR

Septic shock

- Need for vasoactive medication to maintain blood pressure, OR
- Two of the following:
- Unexplained metabolic acidosis (base deficit >5.0 mEq/L)
- Arterial lactate >2 times normal
- Urine output <0.5 mL/kg/hr
- Capillary refill >5 seconds
- Core-to-peripheral temperature gap $>3^{\circ}\text{C}$
- Note: Septic shock involves a complex interaction of distributive, hypovolemic, and cardiogenic components.

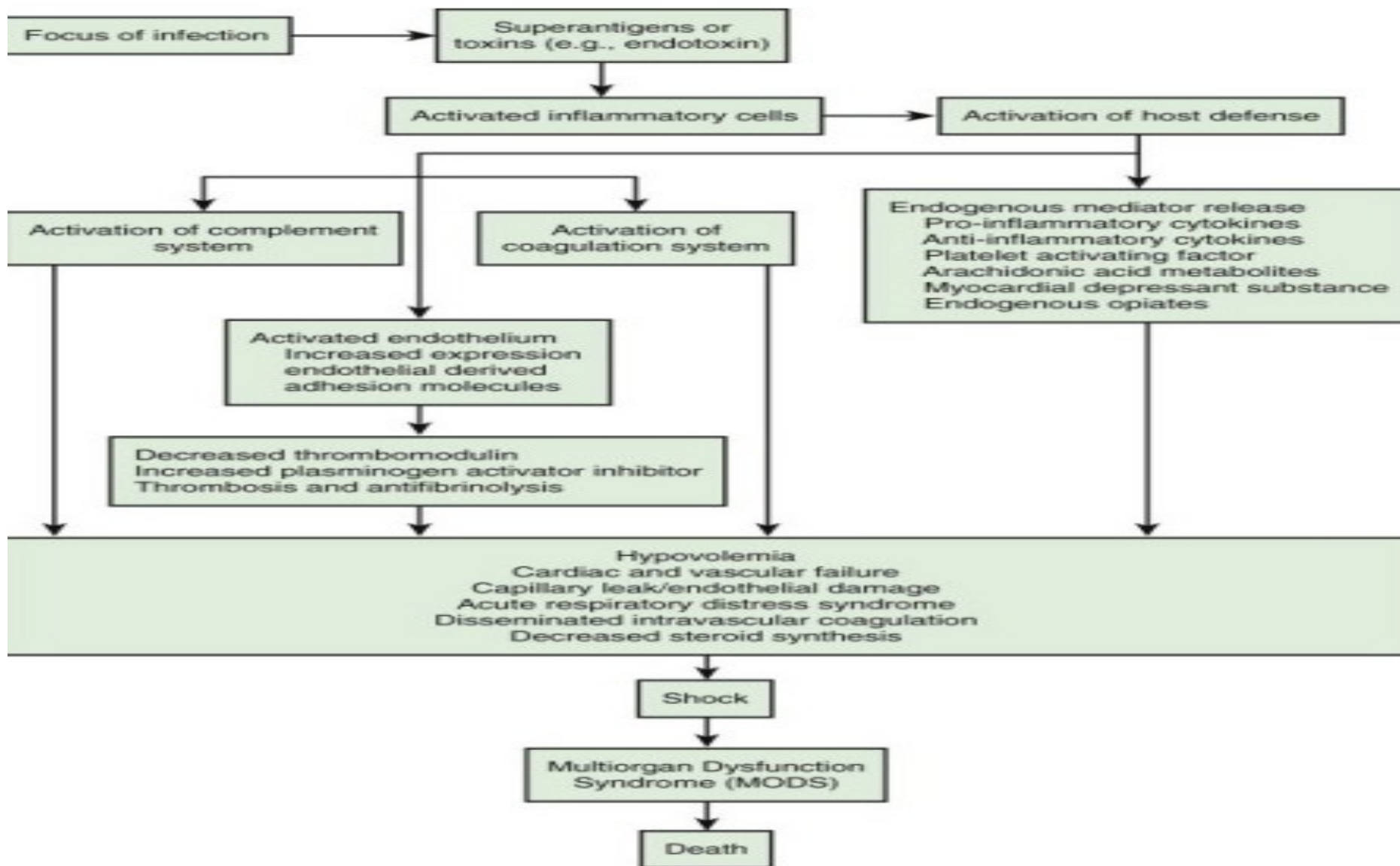


Table 85.2 The Phoenix Sepsis Score^a

VARIABLES	0 POINTS	1 POINT	2 POINTS	3 POINTS
RESPIRATORY, 0–3 POINTS	PaO ₂ :FIO ₂ ≥400 or SpO ₂ :FIO ₂ ≥292 ^b	PaO ₂ :FIO ₂ <400 on any respiratory support or SpO ₂ :FIO ₂ <292 on any respiratory support ^{b,c}	PaO ₂ :FIO ₂ 100–200 and IMV or SpO ₂ :FIO ₂ 148–220 and IMV ^b	PaO ₂ :FIO ₂ <100 and IMV or SpO ₂ :FIO ₂ <148 and IMV ^b
CARDIOVASCULAR, 0–6 POINTS		1 point each (up to 3)	2 points each (up to 6)	
	No vasoactive medications ^d	1 Vasoactive medication ^d	≥2 Vasoactive medications ^d	
	Lactate <5 mmol/L ^e	Lactate 5–10.9 mmol/L ^e	Lactate ≥11 mmol/L ^e	
AGE BASED ^f	Mean Arterial Pressure, mm Hg ^g			
<1 mo	>30	17–30	<17	
1 to 11 mo	>38	25–38	<25	
1 to <2 yr	>43	31–43	<31	
2 to <5 yr	>44	32–44	<32	
5 to <12 yr	>48	36–48	<36	
12 to 17 yr	>51	38–51	<38	

COAGULATION, 0–2 POINTS^H

1 point each (maximum
2 points)

Platelets $\geq 100 \times 10^3/\mu\text{L}$

Platelets $< 100 \times 10^3/\mu\text{L}$

International normalized
ratio ≤ 1.3

International normalized
ratio > 1.3

D-dimer $\leq 2 \text{ mg/L FEU}$

D-dimer $> 2 \text{ mg/L FEU}$

Fibrinogen $\geq 100 \text{ mg/dL}$

Fibrinogen $< 100 \text{ mg/dL}$

NEUROLOGIC, 0–2 POINTS^I

Glasgow Coma Scale score
 > 10 ; pupils reactiveⁱ

Glasgow Coma Scale score
 ≤ 10 ⁱ

Fixed pupils bilaterally

PHOENIX SEPSIS CRITERIA

Sepsis

Suspected infection and
Phoenix Sepsis Score ≥ 2
points

Septic shock

Sepsis with ≥ 1
cardiovascular point(s)

Clinical features
(Based on source of infection)

EARLY

- Poor feeding +/- fever
- A toxic or ill appearance
- Tachycardia, tachypnea,
- Normal capillary refill
- A bulging or sunken fontanelle
- Edema
- Mottled extremities,

LATE

- Respiratory failure,
- Cyanosis,
- Decreased pulses,
- Delayed capillary refill
- Cool extremities,
- Lethargy, and coma

Septic Shock Management

- **First-Hour Bundle (2026 Surviving Sepsis Campaign):**

- 1. Obtain IV/IO Access**

- Two large-bore IVs or IO if access is not achieved within 2 attempts

- 2. Collect Blood Culture**

- Obtain before antibiotics if possible, but do not delay antibiotics

- 3. Administer Empiric Broad-Spectrum Antibiotics**

- **Within 1 hour** of shock recognition
- Choice based on age, immune status, local susceptibility patterns, and source of infection

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- **Neonates (<28 days):** Ampicillin + Ceftazidime (or Cefotaxime) + consider Acyclovir
- **Children >30 days:** Ceftriaxone + Vancomycin
- **Immunocompromised:** Cefepime + Vancomycin
- **Toxic shock:** Add Clindamycin
- **Intra-abdominal source:** Add Metronidazole

4. Measure Lactate

- Baseline lactate helps guide therapy

5. Provide Volume Resuscitation

- 10-20 mL/kg boluses, reassess after each

6. Initiate Vasoactive Medication

- If shock persists despite fluid therapy

Con'd

Clinical Presentations:

- **Cold Shock:** Vasoconstricted, prolonged CRT, weak pulses
→ **Epinephrine**
- **Warm Shock:** Vasodilated, bounding pulses, flash CRT
→ **Norepinephrine**

Shock in Malnutrition

Causes include:-

1. Hypovolemic
2. Septic
3. Cardiogenic

Cont'...

Consider Shock when the child is lethargic or unconscious and has cold hands plus either of the following:-

- Slow capillary refill (longer than 3 seconds), or
- Weak, fast or absent radial or femoral pulses

Cont'...

- Four key steps to manage shock are;
 - Give an oxygen
 - Give 10% glucose
 - Prevent hypothermia
 - Give an Iv fluid and follow up

Septic shock

- Diagnosis has been made by;
 - A sign of hypovolemic shock and
 - Failure to improve after the first hour of IV fluids for management of shock in dehydration

Poor Prognostic Features

Etiology:

- Septic shock carries the highest morbidity and mortality
- Underlying chronic illness

Patient Factors:

- Infant age (highest risk group)
- Pre-existing chronic illness (mortality 6-9% vs. 3% in healthy children)
- Immunocompromised state

Severity:

- Need for mechanical ventilation
- Fluid-refractory shock
- Catecholamine-resistant shock
- Multiple organ dysfunction syndrome (MODS)
- Need for ECMO

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Laboratory Markers:

- Elevated procalcitonin levels (nonsurvivors have significantly higher levels)
- Elevated lactate (persistent hyperlactatemia)
- Progressive metabolic acidosis

Treatment Factors:

- Delayed recognition and therapy
- Delay in antibiotic administration >1 hour
- Fluid overload >10%

Key Message: Shock and MODS remain one of the leading causes of nonaccidental death in infants and children.

Prognosis

- In septic shock, mortality rates are as low as 3% in previously healthy children and 6–9% in children with chronic illness (compared with 25–30% in adults).
- With early recognition and therapy, the mortality rate for pediatric shock continues to improve,

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- But shock and MODS remain one of the leading causes of nonaccidental death in infants and children
- The risk of death involves a complex interaction of factors, including the underlying etiology, presence of chronic illness, host immune response, and timing of recognition and therapy

Treatment Goals and Follow-up

Clinical Goals for Successful Resuscitation:

- **Hemodynamic:**
 - Decreasing heart rate (first sign of reversal)
 - Blood pressure rising to age-appropriate range
 - Strong peripheral pulses
 - Warm extremities
- **Perfusion:**
 - Capillary refill time <2 seconds
 - Improving mental status (less irritable, more alert)

Treatment Goals and Follow-up

Metabolic:

- Urine output >1 mL/kg/hour
- Falling blood lactate levels (≥ 10 -20% decrease)
- Correcting metabolic acidosis

Respiratory:

- Decreasing respiratory rate
- $\text{SpO}_2 \geq 94\%$

Follow-up and Escalation:

- **All shock patients require careful follow-up and monitoring:**
- **Reassessment:** Patients should be reassessed after each intervention

Treatment Goals and Follow-up

- **Transfer:** All patients in shock should be referred to a higher-level facility after emergency care and stabilization
- **Monitor:** Continue to monitor hemodynamic and metabolic parameters
- **Escalation:** If no improvement, consider transfer to PICU or ECMO

Documentation:

- Document response to treatment before and after each intervention
- Monitor trends in vital signs and laboratory values

Summary

- Shock is is life threatening condition if not timely detected and managed.
- We have five general categories of shock
- General treatment measures beginning from ABCD of life
- Treating specific types of shock with
inotropes, antibiotics, blood transfusion in addition to volume expansion.

Con'd

- In fluid-Refractory and Catecholamine-Resistant Shock consideration of other atypical and reversible causes is needed and,
- Second-line Vasoactive Medications, and Stress-dose Corticosteroids can be used for management

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THANK YOU